



TECHNICAL ENGINEERING PAPER

CMERP x GovSec Integration

**Bidirectional interoperability between CMERP (Insight Manager / Sensor Fleet)
and the GovSec Threat Intelligence Platform**

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1. Executive Summary

This integration comprises three deliverables: (1) a GovSec-hosted signature repository generating ET Pro-compatible signatures for CMERP Suricata ingestion; (2) a CMERP Sensor Appliance deployed in the Aras environment; and (3) a Suricata-format TI alert push from CMERP into GovSec.

Deliverable	ID	Direction	Owner	Status
Signature repository (ET Pro-compatible)	COM-20260821-002	GovSec → CMERP	Ahmad Fuad (Aras / GovSec)	Agreed — no external dependency
CMERP Sensor Appliance (Sensor Fleet)	COM-20260821-003	Hosted in Aras env	Tuan Fatah (pending approval)	Subject to management approval (PENDING)
Suricata / TI alert integration	COM-20260821-004	CMERP → GovSec	Tuan Fatah (MyCERT CTRC)	Agreed — JSON param sample in hand

The critical-path blocker is management approval for the CMERP Sensor Appliance deployment (RSK-20260821-001). Deliverable 1, the signature repository, carries no external dependency, and CSM has already provided the alert-parameter JSON sample.

1.1 Present-state vs future-state classification

Per Gate 4 §2: future-state items must not be presented as existing capabilities. The table below classifies each integration component by current state.

Component	State classification	Notes
GovSec TIP	Present-state (MVP)	v3.0 MVP live; signature feed, push ingestion, correlation, PDPA engine operational
LebahNet honeypot network	Present-state (Active)	Ahmad Fuad-confirmed 2026-08-22; CSM national honeypot feeding threat indicators to GovSec via push ingestion (STIX/TAXII 2.1)
CMERP signature feed	Designed / Proposed	ET Pro-compatible output architecture defined; not yet live
CMERP sensor appliance	Pending approval	Deployment subject to management approval (RSK-20260821-001)
Suricata alert integration	Proposed	Flow B contract defined; CSM JSON sample in hand; awaiting sensor deployment
SiberSUITE	Future-state	Separate workstream — telemetry, Score Card, CBOM Agent
CBOM	Future-state	Cyber Bill of Materials — future integration target
Score Card	Future-state	Future-state scoring dashboard — not yet scoped

2. Scope and definitions

2.1 In scope

This paper covers the three CMERP × GovSec deliverables — the signature repository, the sensor appliance, and the alert integration — and adopts the Insight Manager Architecture as the integration blueprint (ART-20260821-003). It describes the bidirectional data flow, in which signatures and IOCs travel from GovSec to CMERP while TI and detection alerts travel from CMERP to GovSec, together with the GovSec entity mapping applied to ingested CMERP alerts. The paper further covers the security, governance, and PDPA controls applicable to the integration, as well as testing, validation, and the phased rollout.

2.2 Out of scope (this paper)

The following matters are excluded from this paper. The GovSec × SiberSUITE integration is a separate workstream covering telemetry, the Score Card, and the CBOM Agent. The AI Co-Design Lab cohort programme is likewise out of scope, as are the GovSec × CMERP commercial terms, which are covered in the collaboration engineering paper, and the VoronDRQ go-to-market and CyberDSA launch logistics.

2.3 Definitions

Term	Definition
CMERP	CyberSecurity Malaysia's Case Management & Enterprise Reporting Platform
Insight Manager	CMERP component that owns the Signature DB, the TI-to-Signature Converter, external IOC ingestion, and versioned ET Pro-compatible rules
Sensor Fleet	CMERP component: Sensor 1/2/3 with detection engines; receives signature updates, emits alert flow
GovSec	Aras's National Cyber Threat Intelligence Platform (TIP) — v3.0 MVP, 4 domains, 12-entity model
ET Pro-compatible	A signature/ruleset output format that GovSec generates, compatible with ET Pro format so CMERP can ingest via Suricata without a new parser
Suricata	The open-source IDS/IPS engine whose alert format CMERP pushes to GovSec
IOC	Indicator of Compromise — IP, domain, file hash, URL, etc.
TI alert	Threat Intelligence alert — a detection event pushed from CMERP to GovSec
Signature DB	The Insight Manager's repository of generated + external signatures, versioned as ET Pro-compatible rules

3. Systems involved

3.1 CMERP — the Insight Manager architecture

Component	Sub-components	Role in integration
Insight Manager	Signature DB; Threat intelligence; TI-to-Signature Converter; Generated signatures / Repository; External signatures (Government: IP, Domain, File hash); Versioned rules and signatures (ET Pro-compatible)	Consumes the GovSec signature feed; converts TI to signatures; ingests external government IOCs; produces versioned ET Pro-compatible rules for the sensor fleet
GovSec (Aras side)	Normalise and generate rules; Signature feed; Elastic (alert index & search, correlation & analysis, detection alerts)	Originates the signature output (ET Pro-compatible); receives and correlates TI alerts from the sensor fleet
Sensor Fleet	Sensor 1, Sensor 2, Sensor 3; Detection engines (x3)	Receives signature updates from the Insight Manager; runs detection; emits alert flow to GovSec Elastic

3.2 GovSec TIP — integration-relevant capabilities (v3.0)

GovSec domain	Capability used by the CMERP integration
Threat Intelligence & Ingestion	Push ingestion API (static API key auth) — candidate channel for CMERP TI-alert push; feed registry + provenance; STIX/TAXII object storage; indicator enrichment + dedup
Analysis & Detection	Dynamic 0–100 threat scoring; correlation engine (shared-ASN, IP/domain resolution, malware family, C2, shared cert); MITRE ATT&CK mapping; CVE/KEV linkage
Alerting & Response Operations	Alert lifecycle (open→ack→investigating→resolved/false_positive); fingerprint dedup; auto-escalation (Critical 15min, High 60min); case management (P1–P4); deterministic case-report generation
Governance & Compliance	PDPA PII engine (NRIC/email/phone/bank/card); Malaysian classification levels (Rahsia→Awam) + compartment access; immutable audit trail; RBAC at route layer

3.3 Hosting boundary

The hosting boundary is drawn as follows: the CMERP Sensor Appliance is hosted in the Aras environment (COM-20260821-003), while the Insight Manager remains CSM-side. The signature repository and GovSec Elastic are both hosted Aras-side.

4. Bidirectional data flow

The integration is bidirectional. Signatures/IOCs flow GovSec → CMERP (outbound). TI/detection alerts flow CMERP → GovSec (inbound). The two flows are decoupled and can be commissioned independently.

4.1 Flow A — Outbound: signatures / IOCs (GovSec → CMERP)

The outbound flow begins when GovSec normalises and generates rules from its indicator store and threat intelligence, then publishes ET Pro-compatible signatures to the Aras-hosted signature repository (COM-20260821-002) for CMERP Suricata ingestion. The CMERP Insight Manager ingests the GovSec ET Pro-compatible output via Suricata into the Signature DB, which is also populated by the Insight Manager's TI-to-Signature Converter and by the external government IOC feed (IP, Domain, File hash). From this combined input, the Signature DB produces versioned rules and signatures in ET Pro-compatible format. The versioned rules are then distributed to the Sensor Fleet as signature updates, and the sensor detection engines apply the updated signatures.

Attribute	Value
Source	GovSec (Aras-side)
Destination	CMERP Insight Manager Signature DB (CSM-side)
Transport	GovSec pushes ET Pro-compatible signatures to CMERP
Format	ET Pro-compatible
Frequency	Versioned; on publish
Auth	Push delivery credentials (TBD — see §6.3)
Deliverable	COM-20260821-002 (Aras-owned, no external dependency)

4.2 Flow B — Inbound: TI / detection alerts (CMERP → GovSec)

In the inbound flow, the sensor detection engines evaluate traffic and configuration against the current signatures, and matching events generate detection alerts in Suricata format. CMERP publishes these alerts to GovSec, which ingests them through the push ingestion API (static API key authentication) into the alert index. GovSec Elastic indexes and searches the alerts, and the correlation engine correlates them with existing indicators, MITRE techniques, CVEs, and APT attribution. The correlated alerts are then available for analyst triage, case creation, and executive reporting.

Attribute	Value
Source	CMERP Sensor Fleet (detection engines) via CMERP
Destination	GovSec Elastic — alert index & search, correlation & analysis
Transport	Push (CMERP pushes to GovSec) — push ingestion API candidate
Format	Suricata (alert); JSON parameter sample already

Attribute	Value
	provided by CSM
Frequency	On detection (event-driven) + batch
Auth	Static API key (per GovSec push ingestion API)
Deliverable	COM-20260821-004 (CMERP-owned; sample in hand de-risks this)

4.3 Flow C — Inbound: LebahNet honeypot indicators (LebahNet → GovSec)

LebahNet is CSM's national honeypot network (confirmed active, Ahmad Fuad-confirmed 2026-08-22). It feeds threat indicators into GovSec via the push ingestion API (STIX/TAXII 2.1, static API key auth). LebahNet is under Tuan Fatah's team (MyCERT CTRC) — the same team as CMERP. Source: OIC-CERT Journal paper 'Malware Discovery using Lebahnet Technology', Vol 2 Issue 1, Feb 2020, authors Fathi Kamil et al. [4]

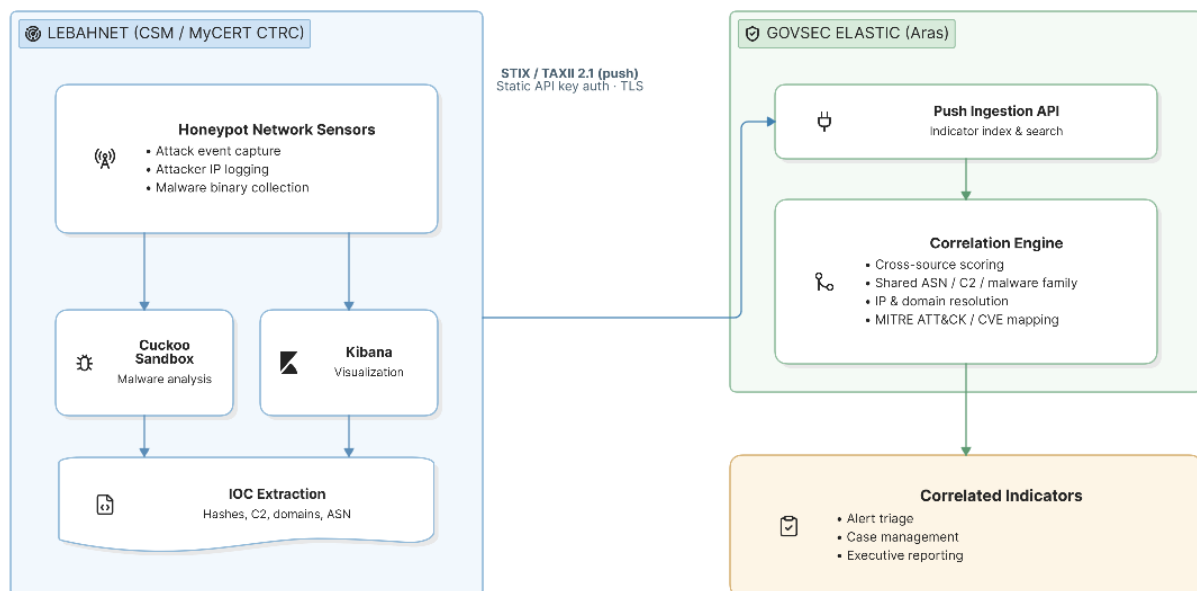
The LebahNet honeypot network captures attack events, attacker IPs, and malware binaries across its deployed honeypot sensors. Captured malware binaries are analysed via Cuckoo Sandbox [5], with the results visualised via Kibana [6]. LebahNet pushes the resulting threat indicators, formatted as STIX/TAXII 2.1, to GovSec through the push ingestion API using static API key authentication. GovSec Elastic indexes the ingested indicators in the alert/indicator index, and the GovSec correlation engine correlates them with existing GovSec indicators through shared-ASN linkage, IP/domain resolution, malware family, C2 infrastructure, and shared certificates. The correlated indicators are then available for analyst triage, case creation, and executive reporting.

Attribute	Value
Source	LebahNet honeypot network (CSM / MyCERT CTRC)
Destination	GovSec Elastic — indicator index & search, correlation & analysis
Transport	Push (LebahNet pushes to GovSec) — push ingestion API (STIX/TAXII 2.1)
Format	STIX/TAXII 2.1
Frequency	On detection (event-driven) + batch
Auth	Static API key (per GovSec push ingestion API)
State	Present-state (Active — Ahmad Fuad-confirmed 2026-08-22)
Owner	Tuan Fatah (MyCERT CTRC)
Reference	OIC-CERT Journal: 'Malware Discovery using Lebahnet Technology', Vol 2 Issue 1, Feb 2020 (Fathi Kamil et al.) [4]

4.3.1 LebahNet component mapping

Component	Sub-components	Role in integration
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Component	Sub-components	Role in integration
LebahNet honeypot network	Honeypot sensors; attack event capture; attacker IP logging; malware binary collection	Captures attack events, attacker IPs, and malware binaries from deployed national honeypots
Cuckoo Sandbox	Malware analysis engine; behavioural analysis; IOC extraction	Analyses captured malware binaries; extracts indicators for feed
Kibana	Visualisation dashboard; threat analytics views	Visualises LebahNet threat data and analysis results
GovSec push ingestion API	STIX/TAXII 2.1 endpoint; static API key auth; indicator storage	Receives LebahNet threat indicators into GovSec Elastic index
GovSec Elastic + correlation	Indicator index; correlation engine; shared-ASN/C2/malware-family linking	Indexes and correlates LebahNet indicators with existing GovSec threat intelligence



Owner: Tuan Fatah (MyCERT CTRC) | State: PRESENT-STATE ACTIVE (Ahmad Fuad-confirmed 2026-08-22)

5. Signature generation & distribution pipeline

5.1 Pipeline stages

Stage	Input	Process	Output	Owner
1. TI ingestion	Threat intelligence (GovSec + external)	Ingest into Insight Manager	Raw TI records	Ahmad Fuad (GovSec TIP)
2. TI-to-Signature conversion	Raw TI records	TI-to-Signature Converter transforms TI into signatures	Generated signatures	Ahmad Fuad (GovSec TIP)
3. External IOC	Government IOCs	External signatures	Merged signature	Ahmad Fuad

Stage	Input	Process	Output	Owner
merge	(IP, Domain, File hash)	feed into Signature DB	set	(GovSec TIP)
4. Signature DB versioning	Generated + external signatures	Signature DB produces versioned rules	Versioned rules & signatures (ET Pro-compatible)	Ahmad Fuad (GovSec TIP)
5. GovSec rule normalisation	GovSec-generated ET Pro-compatible rules	GovSec normalises and generates rules	Normalised signature feed	Ahmad Fuad (GovSec TIP)
6. Sensor distribution	Normalised signature feed	Signature feed → sensor fleet as updates	Updated sensor signatures	Tuan Fatah (MyCERT CTRC)
7. Detection	Updated signatures	Sensor detection engines evaluate	Detection alerts	Tuan Fatah (MyCERT CTRC)
8. Alert return	Detection alerts	Alert flow → GovSec Elastic	Correlated alerts in GovSec	Tuan Fatah (MyCERT CTRC)

5.2 The three Signature DB input sources

Source	Type	Provenance	Owner
TI-to-Signature Converter	Generated signatures	Derived from ingested threat intelligence	Ahmad Fuad (GovSec TIP)
Aras signature repository (COM-002)	GovSec-generated ET Pro-compatible output	GovSec-produced signatures	Ahmad Fuad (GovSec TIP)
External government signatures	IP, Domain, File hash	Government IOC contributions	Ahmad Fuad (GovSec TIP)

5.3 ET Pro-compatible format contract

Each publish under this contract is a versioned ruleset rather than an ad-hoc rules dump. Rule metadata is preserved throughout, including severity, TLP, MITRE technique binding where applicable, and CVE references. Delivery follows a push model in which GovSec pushes ET Pro-compatible signatures to CMERP and CMERP ingests them via Suricata. Provenance is maintained by carrying a `source_feed_id` on every rule, so that cross-source scoring and feed health continue to function in GovSec when alerts return. Finally, consumption is idempotent: CMERP can re-consume a given version without producing duplicate rules.

5.4 Suricata alert format contract (CMERP → GovSec)

The alert fields match the CSM-provided JSON sample, which is the authoritative specification for this leg. Payloads are delivered through the GovSec push ingestion API under static API key authentication, subject to a 10 MB body cap, and are batched. On ingestion, GovSec parses IPs, domains, hashes, and URLs from each alert to create or enrich Indicator entities, and applies fingerprint-based deduplication so that repeated identical alerts do not flood the alert index. Where possible, GovSec also auto-maps alert indicators to MITRE techniques and CVEs.

6. Security, trust & governance

6.1 Trust boundaries

Boundary	Crossing	Control
Aras environment ↔ CSM-side (Insight Manager)	Signature output (ET Pro-compatible)	Push delivery auth; TLS; ET Pro-compatible over authenticated channel
Aras environment (sensor appliance) ↔ GovSec	Alert flow (Suricata/JSON)	Push ingestion API; static API key; TLS; 10 MB cap
CMERP Sensor Appliance ↔ GovSec Elastic	Detection alerts	Push API; API key; TLS
Government IOC feed ↔ Insight Manager	External signatures (IP/Domain/Hash)	Ingestion controls; provenance tagging

6.2 Authentication & transport

For Flow A (signature output), GovSec pushes ET Pro-compatible signatures to CMERP for Suricata ingestion; authentication is via push delivery credentials that remain to be defined, with an API key or mutual TLS recommended for the delivery endpoint. For Flow B (alert push), the GovSec push ingestion API is authenticated by a static API key, which per the GovSec v3.0 API architecture is the only non-JWT endpoint. All transport occurs over TLS, with no plaintext channels permitted. GovSec enforces RBAC and classification at the route layer, and the push ingestion channel is ingestion-only, with no read access.

6.3 Open auth question (flagged)

Open item: signature-feed delivery auth mechanism not yet specified. Aras to propose API key or mutual TLS; CSM to confirm. Contract gap to close in consolidation meeting.

6.4 PDPA & data handling

GovSec's PDPA engine scans ingested alert indicators for Malaysian PII — NRIC, email, phone, bank, and card data — every 4 hours, redacting findings and logging access. By default, data handled through the integration is public, synthetic, or anonymised; sensitive government or incident data in alerts requires explicit approval and handling controls. Retention is governed by configurable policies with a daily purge and per-policy deletion counts. All processing is sovereign: data remains within Malaysian jurisdiction in compliance with PDPA 2010.

6.5 Classification & compartment access

Malaysian classification levels (Rahsia, Sensitif, Terhad, Pertidahan, Awam) apply to ingested alerts. Visibility is compartment-based, meaning analysts see only the alerts of organisations they are authorised to view. Enforcement is performed server-side at the API route layer, with no client-side authorisation decisions.

6.6 Audit

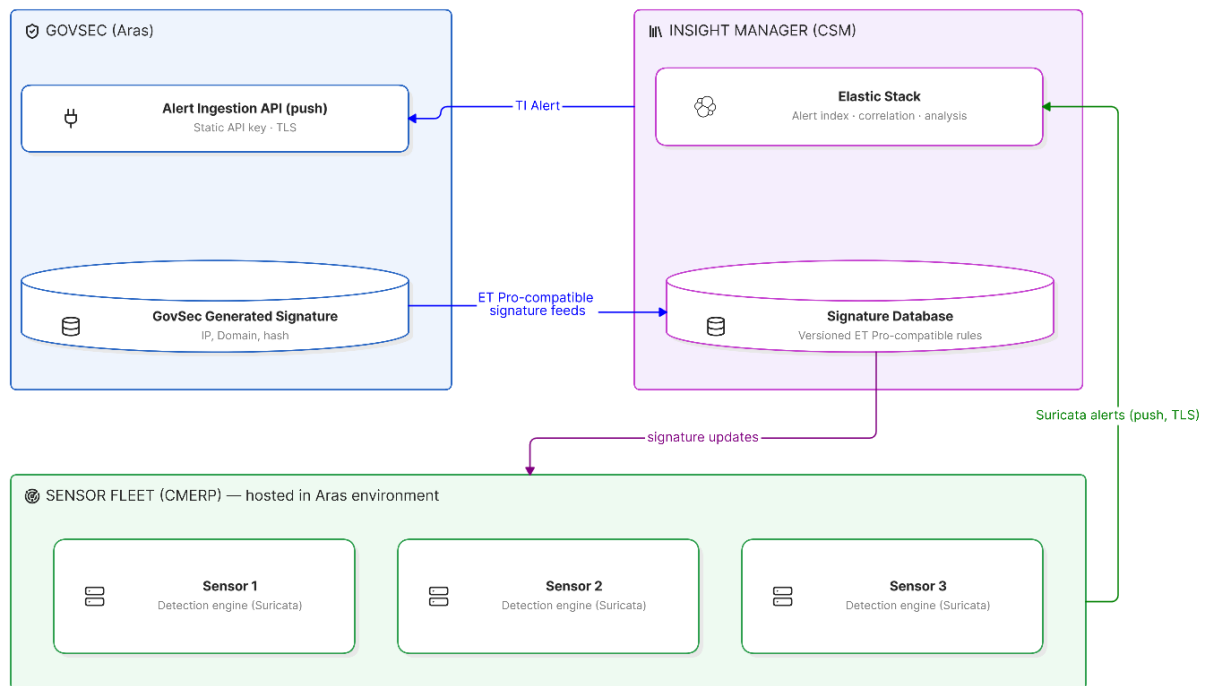
An immutable audit trail records every ingest, alert update, correlation, and sensitive view. Audit events are append-only and attributable to the acting user or service principal, and the push ingestion channel logs every accepted and rejected payload for traceability.

7. GovSec entity mapping for CMERP alerts

CMERP artefact	GovSec entity	Mapping
Suricata alert (JSON)	Alert	Alert record created on ingest; status open; severity from alert; dedup fingerprint computed
IP / domain / hash / URL in alert	Indicator	Indicator created or enriched; provenance = CMERP feed; geo-enriched; severity + TLP assigned
Alert rule reference	Rule	Rule entity referenced; MITRE technique binding where available
Source sensor (Sensor 1/2/3)	Organisation / Feed	Sensor recorded as source feed; provenance preserved (source_feed_id)
Correlated indicators	Correlation	Correlation engine links alert indicators to existing GovSec indicators (shared-ASN, C2, malware family)
MITRE technique (if mapped)	Framework Entity	Auto-mapping pipeline maps indicators to MITRE techniques
CVE reference (if present)	CVE	Indicator linked to CVE; KEV status + CVSS applied to prioritisation
Attribution (if inferable)	Actor	APT attribution pipeline links to national APT library
Affected government org	Organisation	Org-level association; compartment visibility enforced

Dynamic scoring: each CMERP-derived indicator receives a 0–100 composite score (severity baseline + 30-day temporal decay + cross-source boost + correlation density + feed diversity). CMERP-sourced indicators that correlate with existing GovSec indicators score higher.

8. Component topology



8.1 Hosting summary

Component	Hosting	Owner
GovSec (signature feed, Elastic, correlation)	Aras environment	Ahmad Fuad (Aras / GovSec)
Signature repository (ET Pro-compatible)	Aras environment	Ahmad Fuad (Aras / GovSec)
Insight Manager (Signature DB, Converter)	CSM-side	Ahmad Fuad (GovSec TIP)
Sensor Fleet (Sensor 1/2/3)	Aras environment	Tuan Fatah (pending approval)
GovSec push ingestion API	Aras environment	Ahmad Fuad (Aras / GovSec)

9. Testing and validation

9.1 Test strategy

Validate each flow independently, then end-to-end. The JSON alert-parameter sample from CSM is the contract basis for Flow B testing.

Test	Flow	Method	Pass criterion
ET Pro-compatible output contract	A	GovSec generates versioned ET Pro-compatible ruleset; CMERP ingests via Suricata	CMERP Signature DB reflects the versioned rules; no duplicates on re-consume

Test	Flow	Method	Pass criterion
Signature distribution	A	Update signatures; verify sensor fleet receives update	Sensor detection engines load updated rules
Alert contract	B	Push the CSM-provided JSON sample to GovSec push API	GovSec creates an Alert + Indicators; fields match sample
Indicator extraction	B	Push alert with IP/domain/hash	GovSec creates/enriches Indicator entities with provenance = CMERP
Dedup	B	Push identical alert twice	Second push deduped by fingerprint; no duplicate alert
Correlation	B	Push alert whose indicator correlates with existing GovSec indicator	Correlation created; dynamic score recomputed; cross-source boost applied
PDPA scan	B	Push alert containing PII (NRIC/email)	PII detected, redacted, access logged within 4h scan window
End-to-end	A+B	GovSec signature → CMERP → sensor detects → alert → GovSec correlation	Full loop completes; alert correlated and triageable

9.2 Validation gates

Four validation gates govern the integration. Gate 1 covers the Flow A contract and requires that the ET Pro-compatible output be consumable by CMERP; the approver is Ahmad Fuad. Gate 2 covers the Flow B contract and requires that the JSON sample be ingested cleanly by GovSec; the approver is Ahmad Fuad, with the sample already in hand. Gate 3 requires the sensor appliance to be deployed and emitting; the approver is Tuan Fatah, gated on management approval. Gate 4 covers the end-to-end loop; the approver is Ahmad Fuad, with joint sign-off from Tuan Fatah.

9.3 Non-functional validation

Non-functional validation covers four areas. Latency from alert push to indexed alert must fall below a target SLA, to be defined in the consolidation meeting. Throughput is validated through batched alert pushes within the GovSec 10 MB body cap. Availability covers the uptime of the signature delivery endpoint and the push API, supported by retry and backoff on the CMERP side. Security validation applies the OWASP Web and LLM Top 10 baseline per the GovSec v3.0 remediation plan, with TLS verified and an API key rotation process in place.

10. Dependencies and risks

10.1 The critical-path blocker

Management approval for the CMERP Sensor Appliance deployment (COM-20260821-003 / RSK-20260821-001) is the single blocking dependency. Without it, Flow B (alert push) cannot go live because the sensor fleet is not deployed in the Aras environment.

Mitigation proceeds on four fronts. First, the approval is escalated as a critical-path item in the consolidation meeting, for which an ACT-20260820-009 re-proposal is required since the Aug 18/19/20 slots have lapsed. Second, the no-dependency work is sequenced first: Flow A (the signature repository, ET Pro-compatible) starts immediately and de-risks the 28 Aug target. Third, the Flow B contract test harness is built now from the CSM JSON sample, so that ingestion is ready the moment the appliance lands. Finally, if approval slips past approximately 24 Aug, the 28 Aug target is formally re-baselined and CSM (Tuan Fatah) is notified — it must not be allowed to slip silently.

10.2 Risk register (integration-specific)

Risk	Likelihood	Impact	Mitigation
Management approval for sensor appliance not landed in time	Medium	High	Escalate; sequence Flow A first; build Flow B harness from JSON sample; re-baseline if >24 Aug
Signature-feed delivery auth not yet specified	Medium	Medium	Aras proposes API key or mTLS; CSM confirms in consolidation meeting (§6.3)
ET Pro-compatible format compatibility between GovSec output and CMERP Suricata ingestion	Medium	Medium	Contract test at Gate 1; align on versioning, metadata, provenance before live
Alert fields in live push differ from CSM JSON sample	Low	Medium	Contract test harness built from the sample; CSM confirms sample is authoritative
Timeline collision with CyberDSA launch prep (ACT-018, Aug 31) and GPU tokens (ACT-020, Sep 1)	Medium	High	Lock consolidation meeting this week; sequence to avoid Aug 28–Sep 1 contention
PII in CMERP alerts triggers PDPA handling overhead	Low	High	PDPA engine scans every 4h; redaction + access audit; default to non-sensitive test data
Sensor appliance resource contention in Aras environment	Low	Medium	Capacity check before deployment; quotas + monitoring (GovSec GPU Platform Lead pattern)

11. Phased rollout

Three phases; Phases 1 and 2 run in parallel; Phase 3 gated on management approval.

Phase	Scope	Owner	Gate	Target
Phase 1 — Signature repository (Flow A)	Stand up the GovSec signature output in ET Pro-compatible format; CMERP ingests via Suricata; contract test	Ahmad Fuad	Gate 1 — ET Pro-compatible output consumable	Immediate start (no dependency)
Phase 2 — Alert ingestion (Flow B contract)	Build Flow B contract test harness from CSM JSON sample; validate against GovSec push API; indicator extraction + dedup + PDPA	Ahmad Fuad	Gate 2 — JSON sample ingested cleanly	In parallel with Phase 1
Phase 3 — Sensor appliance + end-to-end	Deploy CMERP Sensor Appliance in Aras env; live alert push; end-to-end loop	Tuan Fatah + Ahmad Fuad	Gate 3+4 — appliance live, E2E loop	After management approval; target 28 Aug

12. Open items & contract gaps

#	Open item	Owner to resolve	Where
1	Signature-feed delivery auth mechanism	Ahmad Fuad proposes; Tuan Fatah confirms	§6.3, consolidation meeting
2	Management approval for sensor appliance	Tuan Fatah (CSM management)	RSK-20260821-001
3	Consolidation meeting slot (Aug 18/19/20 lapsed)	Ahmad Fuad re-proposes; Tuan Fatah confirms	ACT-20260820-009
4	Alert SLA (push → indexed)	Ahmad Fuad + Tuan Fatah (joint)	§9.3
5	Confirm CSM JSON sample is authoritative for Flow B	Tuan Fatah (CSM)	§5.4
6	External government IOC feed — who contributes, what format	Tuan Fatah / government partners	§5.2
7	API key rotation process for push ingestion	Ahmad Fuad	§9.3

13. ESF Definition of Done

Per ESF §9 Element 7 (singular named accountability) and Gate 4 critical path (22–27 Aug 2026). The integration is complete when all four validation gates pass with named approver sign-off, evidence on file, and no unresolved failure modes.

13.1 Validation gates

Gate	Gate criteria	Named approver	Date target	Evidence required	Failure mode
Gate 1 — Flow A (signature contract)	ET Pro-compatible output consumable by CMERP Suricata; versioned ruleset; no duplicates on re-consume	Ahmad Fuad	24 Aug 2026	Contract test report; versioned ruleset file; CMERP Signature DB ingest log showing version applied	ET Pro-compatible format mismatch; duplicate rules on re-consume; versioning not preserved
Gate 2 — Flow B (alert contract)	CSM JSON sample ingested cleanly by GovSec push API; Alert + Indicators created; fields match sample	Ahmad Fuad	25 Aug 2026	Alert ingestion log; indicator extraction output; dedup fingerprint verification; PDPA scan result	Alert fields differ from sample; indicator extraction fails; dedup broken; PII not redacted
Gate 3 — Sensor appliance deployed	CMERP Sensor Appliance deployed in Aras env; sensors emitting detection alerts; live alert push confirmed	Tuan Fatah	26 Aug 2026	Deployment confirmation; sensor uptime log; live alert push receipt in GovSec (management approval on file)	Management approval not landed; sensor appliance not deployed; no live alerts emitted
Gate 4 — End-to-end loop	Full loop: GovSec signature → CMERP → sensor detects → alert → GovSec correlation; alert triageable	Ahmad Fuad (joint sign-off with Tuan Fatah)	27 Aug 2026	End-to-end test report; correlated alert in GovSec; triage confirmation; executive report generated	Loop breaks at any stage; alert not correlated; not triageable; scoring not applied

13.2 Success conditions

The integration is DONE when ALL of the following are true:

- Gate 1 passed: GovSec generates versioned ET Pro-compatible rulesets; CMERP ingests via Suricata without errors; re-consume produces no duplicates. Signed off by Ahmad Fuad.
- Gate 2 passed: CSM JSON alert sample ingested via GovSec push ingestion API; Alert entity + Indicator entities created with provenance = CMERP; fields match sample exactly; dedup verified; PDPA scan clean. Signed off by Ahmad Fuad.

- Gate 3 passed: CMERP Sensor Appliance deployed in Aras environment; Sensor 1/2/3 emitting live detection alerts; alerts received by GovSec Elastic. Signed off by Tuan Fatah (management approval on file).
- Gate 4 passed: End-to-end loop complete — GovSec signature → CMERP Suricata → sensor detection → alert push → GovSec correlation; correlated alert is triageable; dynamic score computed; executive report generated. Signed off by Ahmad Fuad + Tuan Fatah.
- All four gates have named approver sign-off, dated evidence on file, and no open failure modes.
- LebahNet Flow C confirmed active: threat indicators flowing from LebahNet to GovSec via STIX/TAXII 2.1 push ingestion; indicators correlated in GovSec Elastic. Confirmed by Tuan Fatah (Ahmad Fuad-confirmed 2026-08-22).

13.3 Failure conditions

The integration FAILS if ANY of the following occurs:

- Management approval for CMERP Sensor Appliance not landed by 26 Aug 2026 — Gate 3 cannot pass; Flow B cannot go live; critical-path blocker (RSK-20260821-001).
- ET Pro-compatible format incompatible between GovSec output and CMERP Suricata ingestion — Gate 1 fails; contract test fails.
- CSM JSON alert sample fields do not match live push — Gate 2 fails; alert contract invalid.
- Signature-feed delivery auth mechanism not specified/agreed — Flow A cannot go live; open item §6.3 unresolved.
- Consolidation meeting not scheduled this week (Aug 18/19/20 slots lapsed) — contract gaps remain open; timeline at risk.
- LebahNet push ingestion to GovSec not functioning (STIX/TAXII 2.1 endpoint unreachable or auth failure) — Flow C degraded despite present-state active status.
- Any gate lacks named approver sign-off or dated evidence — per ESF §9 Element 7, unsigned gates do not count as passed.
- Timeline slips past 27 Aug 2026 without formal re-baseline — silent slip is a failure condition (per §10.1 mitigation).

13.4 Gate 4 critical path alignment (22–27 Aug 2026)

Date	Critical-path milestone	Owner	Gate
22 Aug 2026	Consolidation meeting; close open items §6.3 (auth), SLA, IOC feed	Ahmad Fuad	—
24 Aug 2026	Flow A contract test (ET Pro-compatible output → CMERP ingest)	Ahmad Fuad	Gate 1
25 Aug 2026	Flow B contract test (JSON sample → GovSec push API)	Ahmad Fuad	Gate 2
26 Aug 2026	Sensor appliance deployed (if approval landed); live alert push	Tuan Fatah	Gate 3

Date	Critical-path milestone	Owner	Gate
27 Aug 2026	End-to-end loop test; all gates sign-off; document closure	Ahmad Fuad + Tuan Fatah	Gate 4

14. Decisions requested (CMERP × GovSec only)

1. Approve the Insight Manager integration architecture (Section 4) and the three-deliverable scope.
2. Escalate and confirm management approval for the CMERP Sensor Appliance deployment (RSK-20260821-001) — the critical-path blocker.
3. Confirm the CSM-provided JSON alert-parameter sample is the authoritative contract for Flow B (alert push).
4. Confirm the signature-feed delivery auth mechanism (Aras proposal: API key or mTLS).
5. Confirm the consolidation meeting slot this week (Aug 18/19/20 lapsed — re-proposal required).
6. Identify the external government IOC feed contributors and format (Section 5.2).
7. Endorse the phased rollout (Section 11): Flow A and Flow B contract work start immediately; Phase 3 gated on approval.

15. Appendix — provenance

15.1 Source records

- ART-20260821-003 — Insight Manager Architecture diagram (authoritative blueprint; OCR-extracted)
- CONV-20260821-002 — GovSec × CMERP integration continuation (deliverables, timeline, dependencies)
- CONV-20260821-003 — CMERP data flow recap (signature repo, ET Pro-compatible output, TI alert push, JSON sample)
- COM-20260821-002/003/004 — the three deliverable commitments
- RSK-20260821-001 — management approval dependency + timeline collision risk
- STK-20260821-003 — Tuan Fatah (MyCERT/CSM team lead)
- INIT-20260821-001 — GovSec × CMERP interoperability integration
- ACT-20260820-009 / ACT-20260820-010 — session date + review materials

15.2 GovSec TIP v3.0 references (read-only, upstream)

- GovSec TIP MVP Specification v3.0 (DOC-20260814-003) — entity model, push ingestion API, PDPA engine, correlation, scoring, RBAC/classification
- BRIEF-20260817-001 — CSM Workstream Analysis (WS2 context)

16. References

8. OASIS, "Structured Threat Information Expression (STIX) v2.1," OASIS Standard, 2021. [Online]. Available: <https://docs.oasis.org/open/cti/stix/v2.1/stix-v2.1.html>
9. OASIS, "Trusted Automated Exchange of Intelligence Information (TAXII) v2.1," OASIS Standard, 2021. [Online]. Available: <https://docs.oasis.org/open/cti/taxii/v2.1/taxii-v2.1.html>
10. Open Information Security Foundation, "Suricata User Guide," [Online]. Available: <https://docs.suricata.io/>
11. F. K. Mohad Zainudin, I. H. Ishak, S. Sulaman, F. Ramlee, N. S. Jamaludin, and S. Chantando, "Malware Discovery using Lebahnet Technology," OIC-CERT Journal of Cyber Security, vol. 2, no. 1, Feb. 2020. [Online]. Available: <https://www.oic-cert.org/en/journal/vol-2-issue-1/malware-discovery-using-lebahnet-technol.html>
12. Cuckoo Sandbox Foundation, "Cuckoo Sandbox Documentation," [Online]. Available: <https://cuckoo.readthedocs.io/>
13. Elastic N.V., "Kibana Documentation," [Online]. Available: <https://www.elastic.co/guide/en/kibana/current/>

Document Acknowledgement & Authorisation

This document is acknowledged as a truthful and accurate representation of the CMERP × GovSec integration technical architecture, data flows, and engineering specifications as of August 2026.

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